THE UNIVERSITY OF AIZU 会津大学

School of Computer Science and Engineering

INTRODUCTION TO DATA MANAGEMENT

Foundation Concepts

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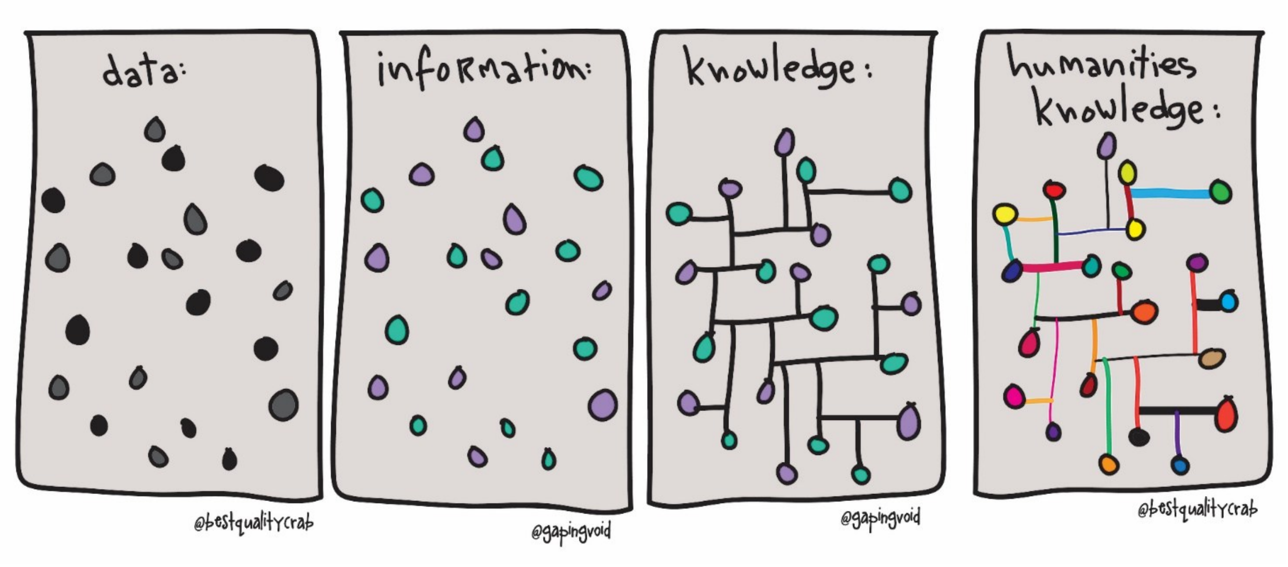
Arts ▪ Languages ▪ Digital Humanities

https://figshare.com/articles/data_information_knowledge_humanities/1397453

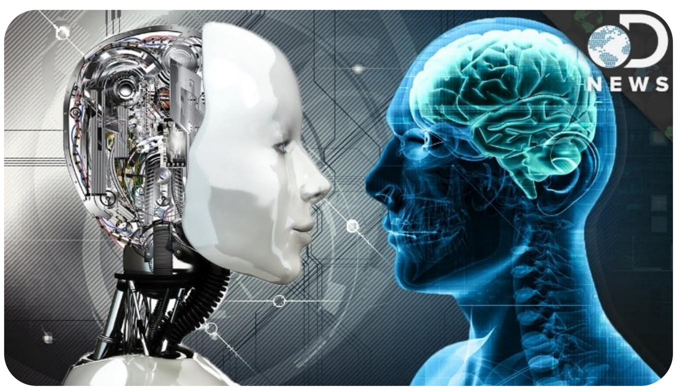


Unit 1

INFORMATION, DATA, KNOWLEDGE

Information and Data

- Information-centric analysis
  - Data provides a way to represent information
- Data-centric analysis
  - Information is a “managed” data



Information	Data
Required by human	Processed by computers
Data structured and presented within a context	Raw and unorganized
Information retrieval	Data mining

What is Information?

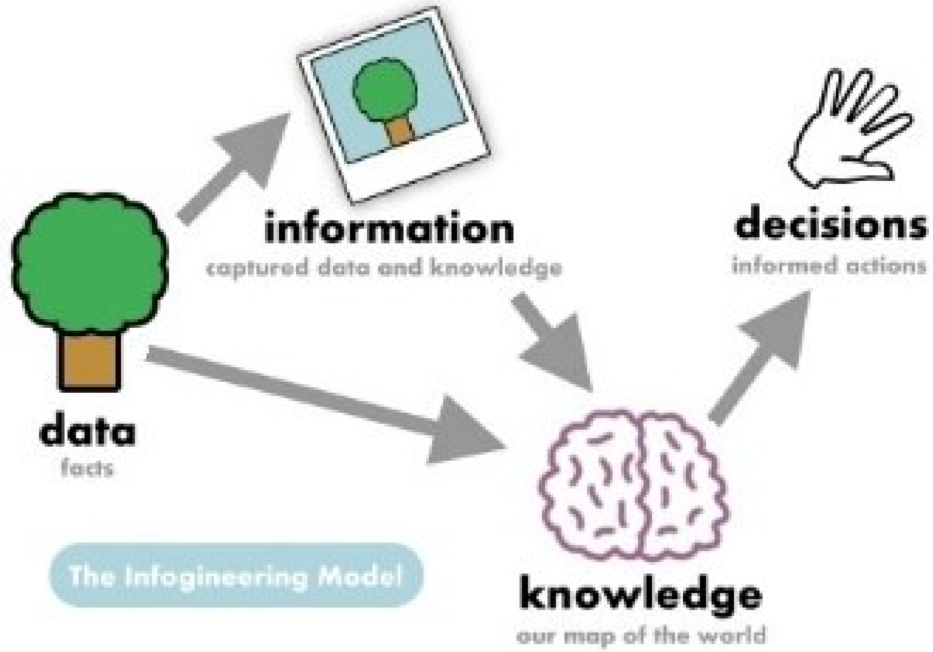
- Knowledge obtained from investigation, study, or instruction
  - Knowledge about natural phenomena
  - Facts about events
  - Data existing in technical devices
  - More
- **Data** represent information in a material form

VERY SIMPLIFIED EXAMPLE

Data: I have one item. *The data displays a 1, not a zero.*
Information: It's a tomato. *Now, we understand the item and its characteristics.*
Knowledge: A tomato is a fruit. *We can identify patterns in the information and apply them to the item.*
Wisdom: Tomato is never added to a fruit salad. *There is an underlying, commonly understood principle that governs the item's purpose.*

https://www.business2community.com/strategy/difference-data-information-0967136

Information, Data, Knowledge, Decisions



http://www.infogineering.net/data-information-knowledge.htm

- When people confuse data with information, they can make critical mistakes
  - Data is assumed to be always correct
    - I can't be 29 years old and 62 years old at the same time
  - Information can be wrong
    - There could be two files on me, one saying I was born in 1981, and one saying I was born in 1948. Both are wrong ☺
- Information captures data at a single point
- The data changes over time
- The information people are looking at is not always an accurate reflection of the data
- ***By understanding the differences between information and data, you can better understand how to make better decisions based on the accurate facts.***

One Very Good Explanation

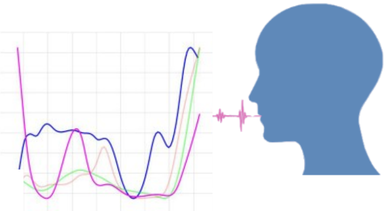
PLEASE LISTEN TO THIS STORY ATTENTIVELY!

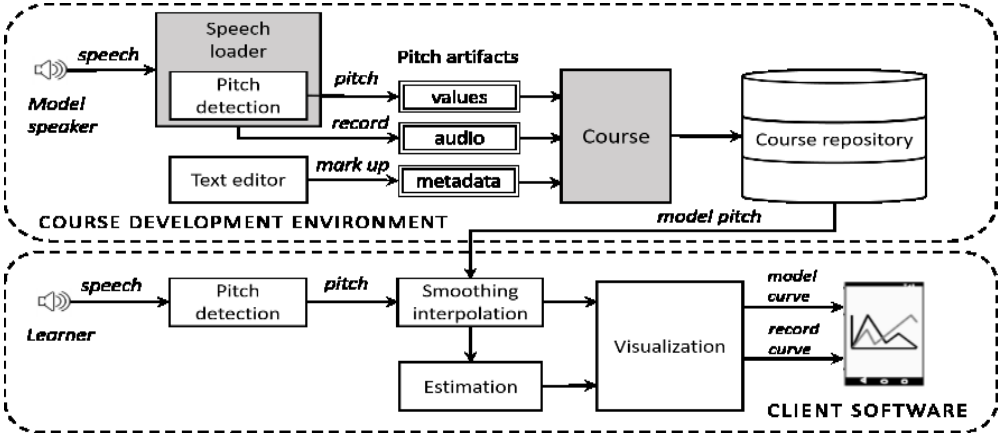
https://www.youtube.com/watch?v=sdzUfHwNCVQ

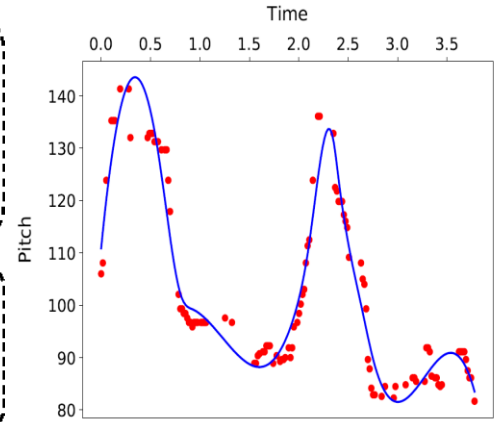
- Data
  - Time
  - Pipe pressure measurements
- Information
  - Graph shown how the pressure changes
- Knowledge
  - What is the normal pressure
  - What to do in order to control the pressure

Some more home reading
https://keydifferences.com/difference-between-data-and-information.html

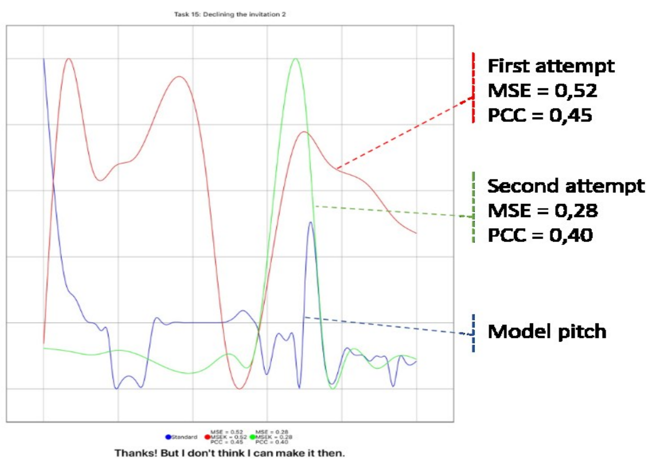
Another Story: Our Ongoing Project

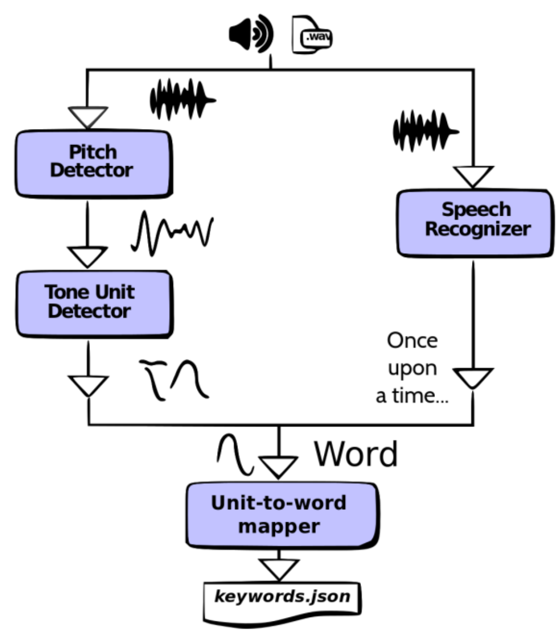






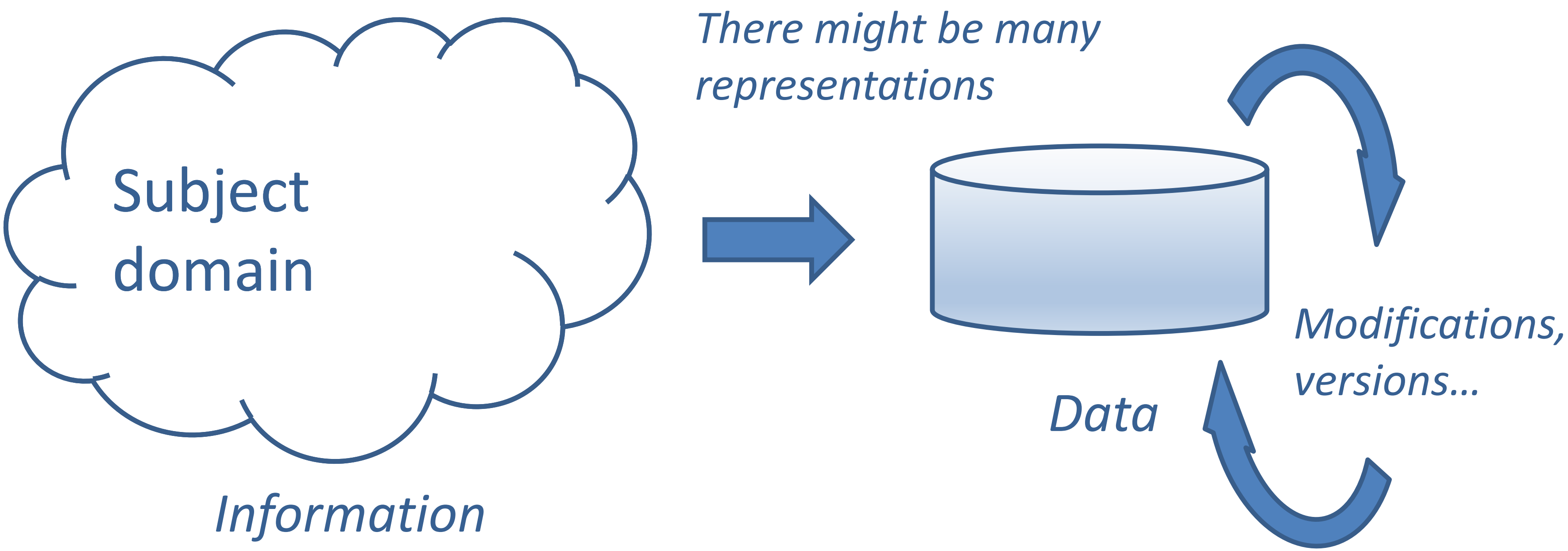
- Data
  - Pitch time series
  - Measurements of difference between the master and user pitches
- Information
  - Approximated curves showing how the intonation changes
  - Pitch quality evaluation
- Knowledge
  - What to do to improve your pronunciation
  - How to use post-processed prosody signals to extract additional information from unstructured speech data





What Do We Mean by Data? (Technically)

- **Data**
  - Facts that can be analyzed or used (e.g. in computers) in an effort to gain knowledge or make decisions
  - Information represented in some form suitable for acquisition, interpretation, storage, and processing

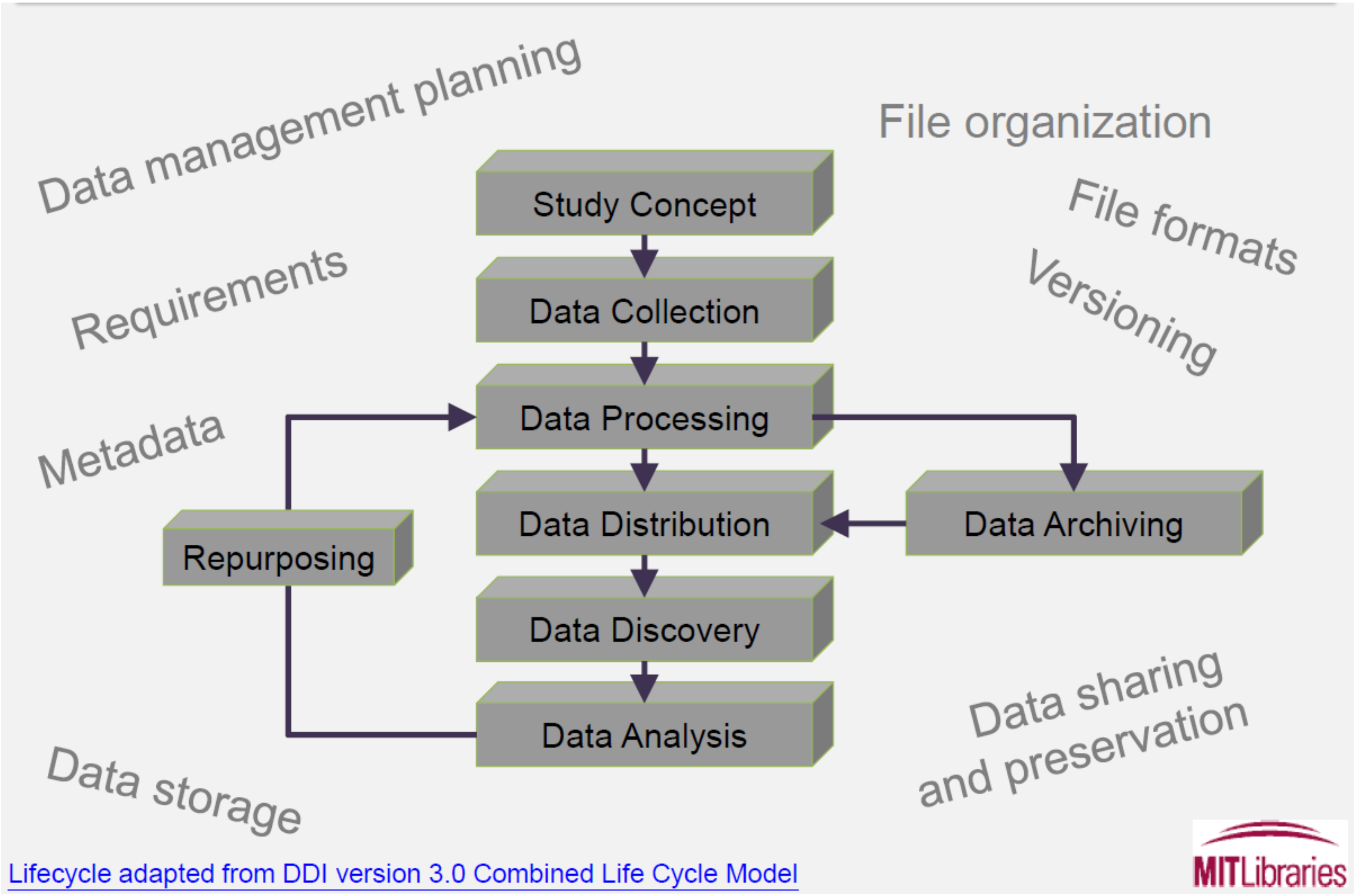


What Do We Mean by Data (Substantially)

General	Social Sciences	Natural/Physical Sciences
Images	Surveys and Polls	Observations and Field Studies
Audio	Focus Group and Individual Interview Transcripts	Sensor and Lab Measurements
Video	Economic Indicators	Computer Modeling
GIS Data	Demographics	Simulations
Numerical Measurements	Social Networking Data	Specimen

https://www.freeiconspng.com

Research Data Lifecycle*



* *MIT OpenCourseWare RES.STR-002 Data Management Spring 2016*

Important Questions

- What are your goals?
- What are you collecting?
- What are you keeping?
- Where do you want to keep it?
- What do you need to be able to use it, to change it, to share it or to repurpose it later?

VERSIONING
VERSIONING
VERSIONING

S
T
O
R
A
G
E

BACKINGUP

COMPRESSION

NCRYPTN

(Re)presentation

http://digitalpublishing101.com/how-to-use-metadata-to-sell-ebooks/

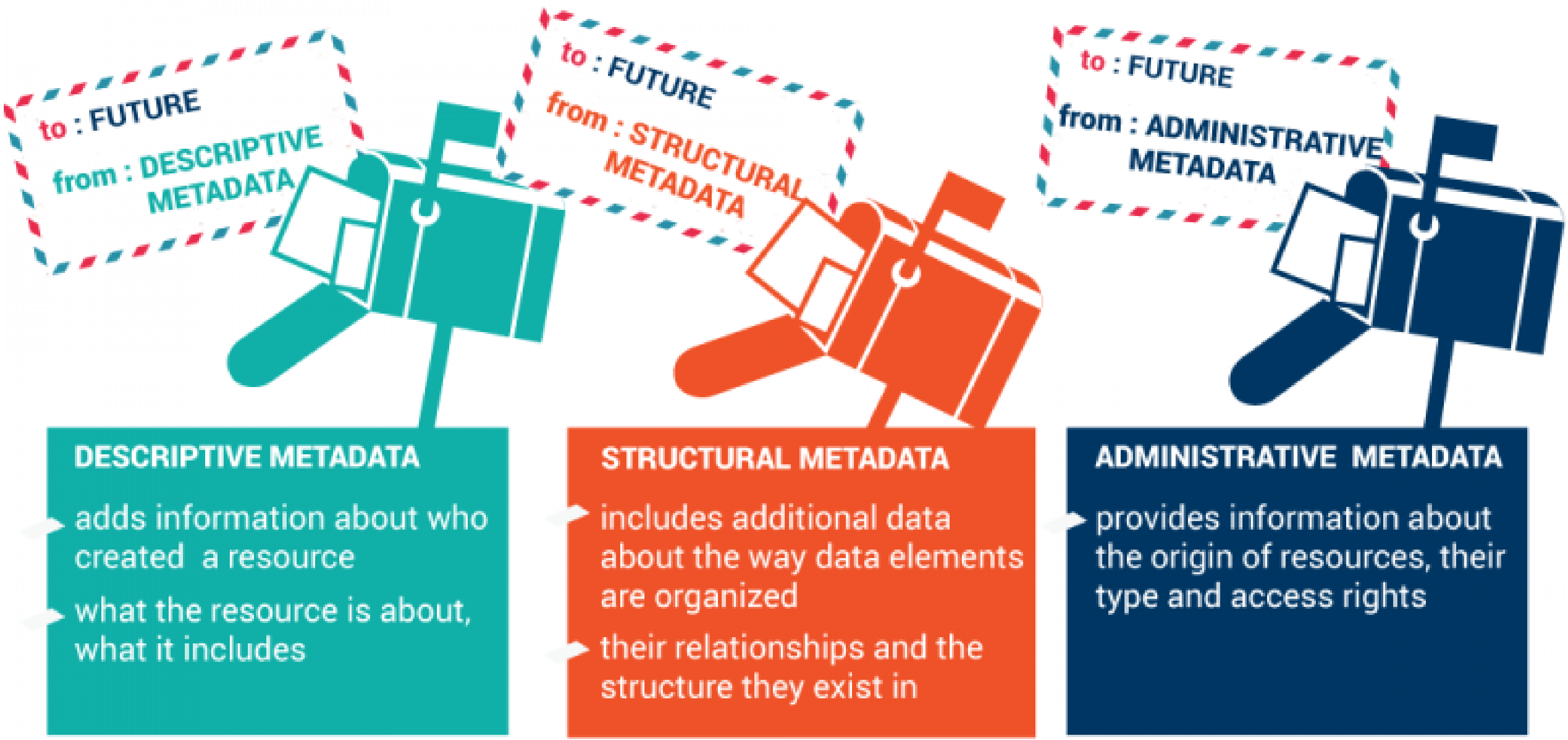
Unit 2

DATA AND METADATA

Data vs Metadata

- Metadata: Why does it matter?
  - Data is not self-describing
  - Metadata, or “data about data” explains your dataset and allows you to document important information for:
    - Finding the data later
    - Understanding what the data is later
    - Sharing the data (both with collaborators and future secondary data users)
    - Consider it an investment of time that will save you trouble later several-fold

Types of Metadata



https://ontotext.com/knowledgehub/fundamentals/metadata-fundamental/

Metadata Standards*

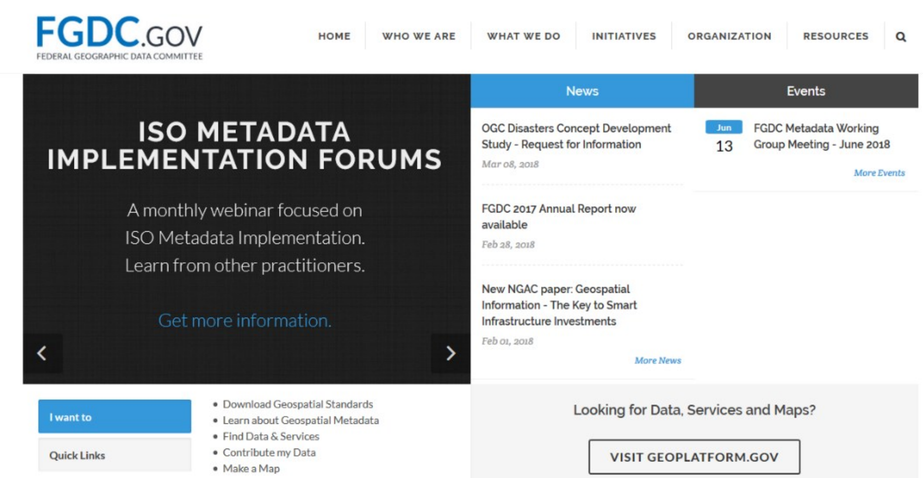
- FGDC (Federal Geographic Data Committee)
- DDI (Data Documentation Initiative)
- Dublin Core (Resources Description)
- Darwin Core (Biodiversity Informatics)
- ABCD (Access to Biological Collections Data)
- AVMS (Astronomy Visualization Metadata Standard)
- CSDGM (Content Standard for Digital Geospatial Metadata)

- Advantages:
  - Ensure you have a complete, standard set of information about each part of your data
  - Enable your dataset to be organized with other datasets

* MIT OpenCourseWare RES.STR-002 Data Management Spring 2016

Metadata Examples: FGDC

- Federal Geographic Data Committee
  - Geospatial metadata
- Geospatial metadata describes maps, Geographic Information Systems (GIS) files, and other location-based data resources



Some more home reading
https://www.fgdc.gov/metadata/documents/WhatIsMetaFiles/WhatIsMetadataPDF

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<?xml version="1.0"?>
 <metadata>
   <idinfo>
     <citation>
       <citeinfo>
         <origin>UoA</origin>
         <pubdate>20180518</pubdate>
         <title>Custom Link Sample</title>
         <geoform>tabular digital data</geoform>
         <pubinfo>
           <pubplace>Aizu-Wakamatsu, Japan</pubplace>
           <publish>Aizu</publish>
          </pubinfo>
          <onlink>http://www.u-aizu.ac.jp</onlink>
        </citeinfo>
       </citation>
        ....
```

Metadata Examples: DDI

- Data Documentations Initiative
  - An international standard for describing the data produced by surveys and other observational methods

- Social, behavioral, economic, and health sciences

Codebook Samples
http://www.ddialliance.org/resources/markup-examples/codebooks

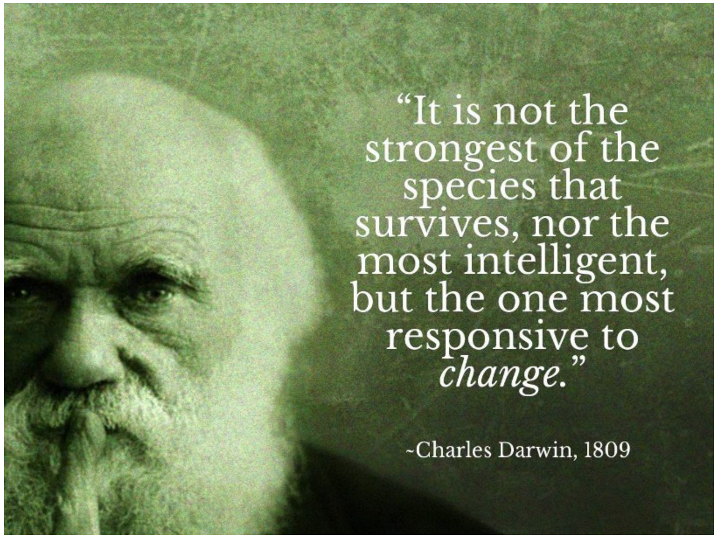
Metadata Examples:
Dublin Core & Darwin Core

- Dublin Core Schema
  - Set of vocabulary terms that can be used to describe:
    - digital resources (video, images, web pages, etc.), physical resources (books or other media), and
    - objects like artworks
  - *«broad and generic being usable for describing a wide range of resources»**

**** The Elements***
http://dublincore.org/documents/dces/

- Darwin Core
  - an extension of Dublin Core for **biodiversity informatics**

the application of informatics techniques to biodiversity information for improved management, presentation, discovery, exploration and analysis



Managing (Research) Data*

- Things you want to check and re-check over time
  - Is the data still **stable and retrievable**?
  - Is the metadata still **available and understandable**?
  - Are the formats still **usable**?

  - Is the **software still available**?
  - Is any **specialized hardware still available**?

  - Is the data still in the **correct location**?
  - Are my **backups working** as I expect?

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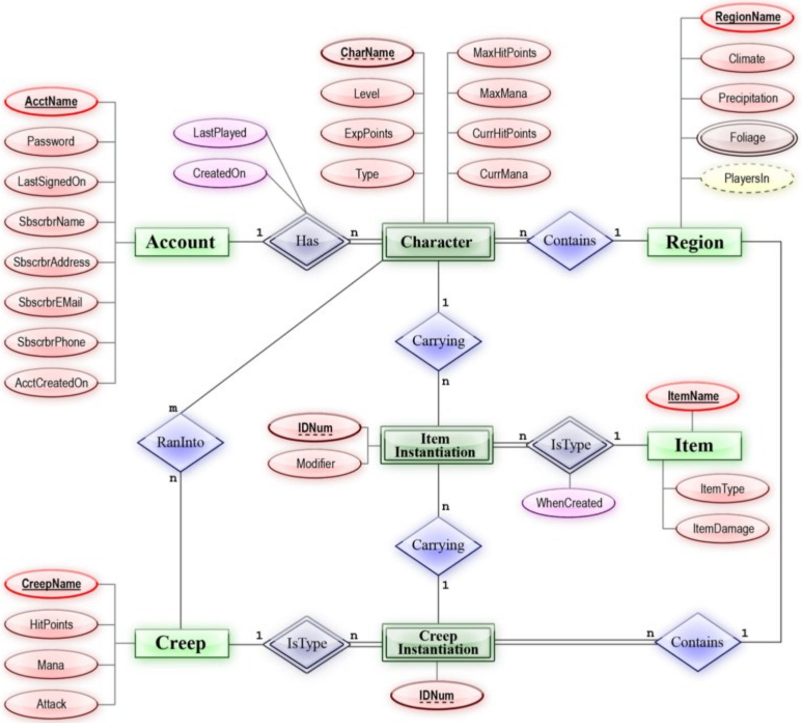
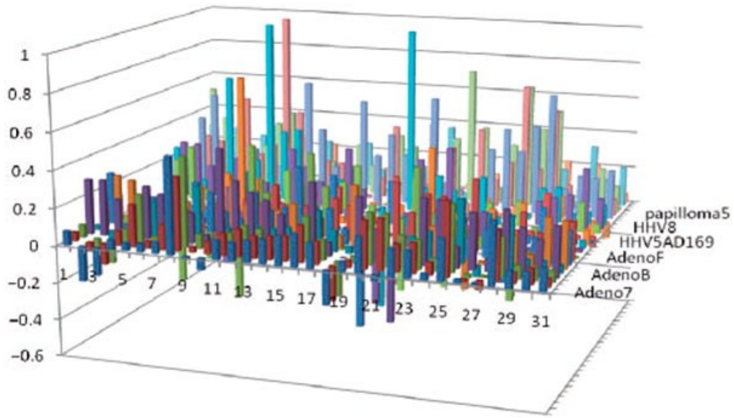


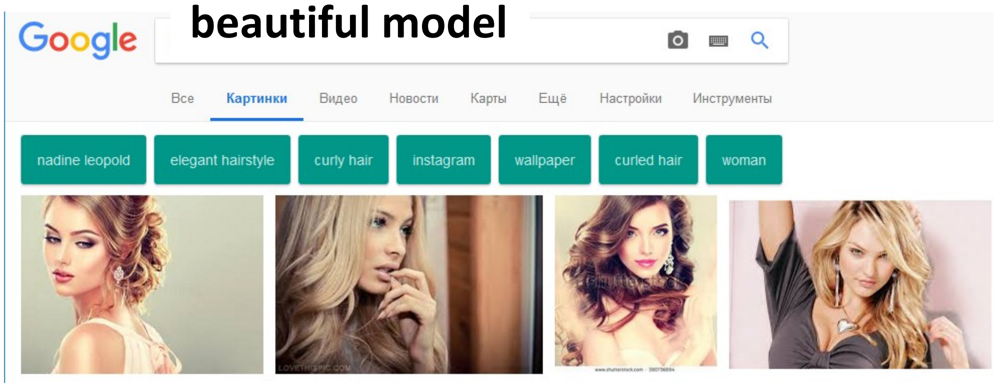
https://en.wikipedia.org/wiki/Entity–relationship_model

Unit 3

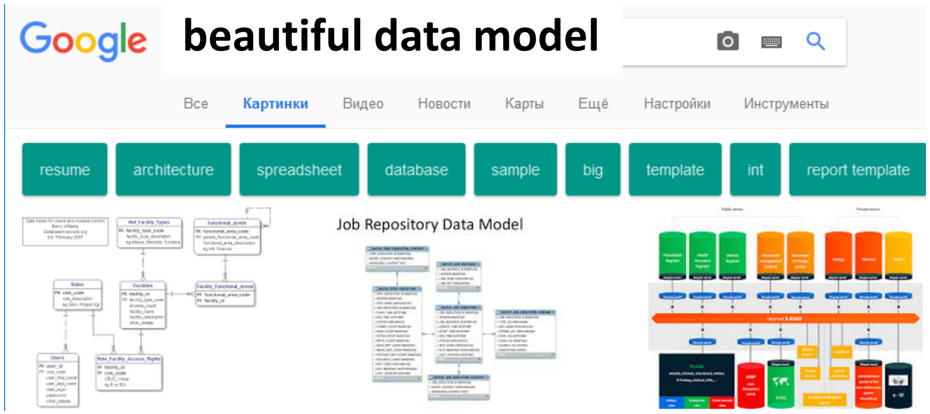
INTRO TO ENTITY RELATIONSHIP MODEL

Data Models Are Not (Only) About Making Ugly* Data Looking Beautiful ☺





- **Data model is a number of concepts used to describe data, the ways to process them, the relationships between them as well as constraints applicable to them**
- Data model defines
  - Data structures
  - Data relationships
  - Constraints (e.g. relationship cardinality)
  - Operations



** Most Ugly & Useless Infographic Competition: The Winners, http://infosthetics.com/archives/2009/12/most_ugly_useless_infographic_the_winners.html*

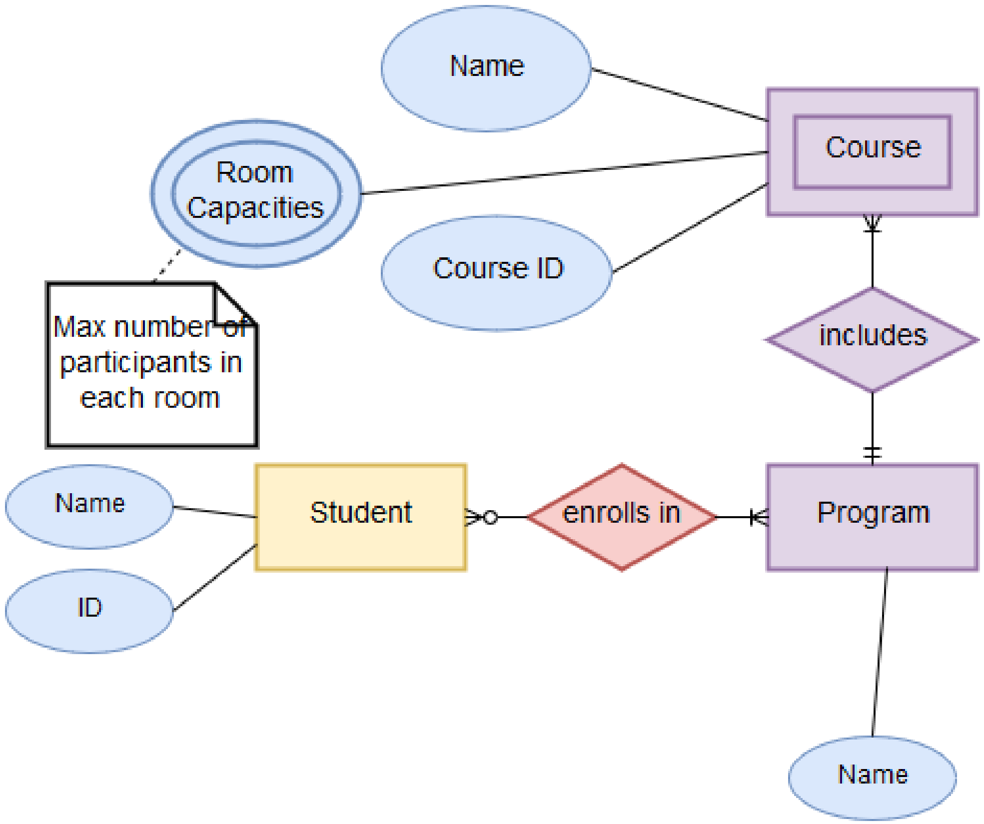
Entity-Relationship Model

- ER Model* (Peter Chen, 1976)
  - describes interrelated things of interest in a specific domain of knowledge;
  - represent things that a business needs to remember in order to perform business processes;
  - can be used also in the specification of domain-specific ontologies.
- "*The entity-relationship model adopts the more natural view that the real world consists of entities and relationships. It incorporates some of the important semantic information about the real world.*“*

* Chen, Peter (March 1976). **"The Entity-Relationship Model - Toward a Unified View of Data".** ACM Transactions on Database Systems. 1 (1): 9–36. doi:10.1145/320434.320440

Example: Academic Programs (Preparation for Exercise)

- The university offers one or more programs.
- A program is made up of one or more courses (at least one).
- A student must enroll in one program.
- The courses are parts of the program.
- A program has the attributes:
  - Name
- A course has the attributes:
  - Name
  - Capacity
  - Course ID
- A Student has the attributes:
  - Name
  - ID



Diagraming Tool

- We use app.diagrams.net (former draw.io)
  - Online, but you can save your diagrams
  - Easy to learn
  - Supports ER diagrams, but other diagrams as well
  - Supports export to other formats
- A couple of useful videos
  - » Quick start
  - » Tutorial

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Foundation Concepts

Q & (maybe) A